

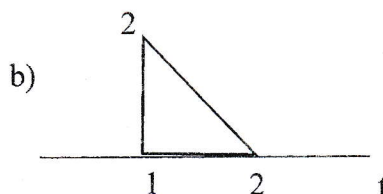
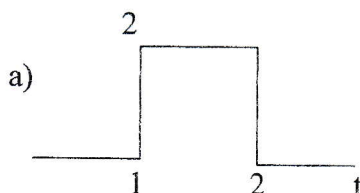
Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEI	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Communication Systems (EX 656)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.



- Sketch and describe the functional digital communication system working in a duplex mode and contrast it with an analog communication system. Mention its impairments and performance parameters. [5+3]
- Determine whether the following signals are: [4+4]
 - $y(t) = 2t^2x(t) + 3tx(t-4)$ linear or not
 - $y(t) = x^2(t) + x(t+2)$ causal or not
- Derive auto-correlation, cross-correlation and convolution of the following two signals graphically: [3+3+3]



- Consider that the bit sequence 100000000001101 is to be transmitted. Draw the resulting waveform if the sequence is transmitted using Polar RZ, Differential Manchester, HDB3 and B8ZS, and compare their bandwidths. [10]
- Derive the expression for figure of merit for the SSB receiver. Define an efficient constellation diagram of a 32-QAM. [5+3]
- A (6, 3) error control code has the following parity check matrix. [2+4+4]

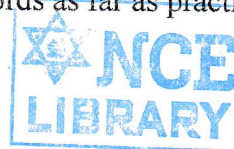
$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$
 - Determine the generator matrix G
 - Generate a codeword for message 110
 - Check whether the received code word 110111 as has error or not.
- Encode "Engineer Ko Betha Arulai Ke Thaha" using Huffman Encoder and calculate the transmission efficiency. Find the codeword if the message is received as "Engineer Ko Betha Ke Ke". [7+2+3]
- Write short notes on: (Any Three) [3x5]
 - Hilbert transform
 - T1 and E1 hierarchy used in telephony
 - Pre-Emphasis and De-Emphasis
 - Trellis diagram of a message 1010 using a $\frac{1}{2}$ rate 3-state memory block convolutional encoder
 - FDM vs TDM

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1. With a suitable example explain Shannon-Hartley channel capacity theorem. Determine whether the given signal $x(t) = e^{-2t}u(t)$ is a power or an energy signal. [3+5]
2. Determine whether the following signals are [3+3]
 - a) $y(t) = x^2(t)$ linear or not.
 - b) $y(t) = y(t-4) + x(t-4)$ time invariant or not
3. a) Derive the expression of double-tone AM, and define BW and modulation indices. [5]
 b) For the given modulated signal draw line spectrums.
 $s(t) = 50\cos(2\pi 10^6 t + 60^\circ) - 15\cos(2\pi 10^6 t - 60^\circ) + 20\sin(4\pi 10^4 t - 190^\circ) + \sin(2\pi 10^4 t + 190^\circ)$. [3]
4. a) Explain the Armstrong's method for Frequency Modulation. [4]
 b) Why do we need pre-emphasis and de-emphasis circuits in FM? Explain. [4]
5. Encode the message "Mississippi is missing" using weighted Huffman code and calculate transmission efficiency. [8]
6. Explain the Delta and Adoptive Delta Modulations encoders and decoders with their derivations and diagram. List out their merits and demerits. [6+2]
7. a) Represent the given binary sequence 110011000001 in unipolar RZ, Manchester and HDB3 encoders. [6]
 b) Explain the modulation scheme for MSK with appropriate example. [4]
8. Define optimum detector and find the impulse response of optimum detector in the presence of additive white noise. Briefly explain Hilbert Transform. [2+6+2]
9. What is the significance of constellation diagram? Compare E1 and T1 TDM PCM hierarchy. [2+4]
10. Determine the encoded sequence for the following input message
 $(m_0, m_1, m_2, m_3, m_4) = (1, 0, 0, 1, 1)$ using convolutional encoder having shift registers with three flip flops. Draw state and trellis diagrams. [8]

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1. What are the reasons for modulation? Write the advantages of digital communication over analog communication. Sketch a generic block diagram of a digital communication for full-duplex mode. [2+3+5]
2. Represent Unit step signal in terms of Signum function. Also, determine whether a Unit step signal is energy or power type or neither of the two. [4+4]
3. Derive the expression for double tone Am. How DSB is different from SSB signal? [6+2]
4. What is the aim of source coding? Encode "Kun Mandir Ma Janchhau Yatri" using Huffman codes and finds its efficiency. [10]
5. Compare Pulse Code Modulation (PCM), Differential Pulse Code Modulation (DPCM) and Delta Modulation. Find the Signal to Quantization Noise ratio (SQNR) of Pulse Code Modulation (PCM). [3+5]
6. Given the binary sequence 1101010111 represent in Unipolar RZ, Bipolar NRZ, Polar NRZ and Manchester encoders. Explain communication impairments with examples. [6+2]
7. What do you mean by optimum detector? Show that the impulse response of the matched filter is reverse delayed version of the input signal. [2+6]
8. Compare TDM and FDM. Show that for voice application. Compare E1 and T1 hierarchies. [3+2+5]
9. Differentiate error-detection and error-correction. Design a convolutional encoder having code-rate of $\frac{1}{2}$. Also, draw the code-tree and trellis diagram for the same assuming any three-bit input. [2+4+4]

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1. Differentiate among noise, interference and distortion. [5]
2. Describe Hilbert Transformation and its properties. Compute the energy and power of unit step signal. [2+5]
3. List out any three properties of autocorrelation(AC) function. Mention the autocorrelation function of white noise. [3+5]
4. a) Why SSB modulation scheme is preferred over DSB, DSB-SC modulation schemes? The total content of an AM signal is 1000w, Determine the power being transmitted if carrier frequency and at each sideband when modulation percentage is 100%. [2+3]
- b) Explain QAM modulation and demodulation with its required diagram. [5]
5. What is the relation between psdf and Autocorrelation function? Explain the Stereo FM encoder and decoder with spectral diagram. [3+7]
6. Explain the aperture effect during flat-topped sampling. Illustrate, the DPCM scheme that overcomes the disadvantages of PCM. A delta modulator system is designed to operate at 5 times the Nyquist rate for a signal having a bandwidth equal to 3kHz bandwidth. Calculate the maximum amplitude of a 2kHz sinusoidal for which the delta modulator does not have slope overload. The given step size is 250 mv. [2+3+4]
7. Represent 100111010 using following encoders. [2+2+2+2]
- a) Polar RZ b) Bipolar NRZ c) AMI d) Manchester
8. What are the significances of multilevel modulation? Explain QPSK with its transformation as well as a receiver block diagram. [2+4]
9. a) Explain BPSK modulation technique with its relevant diagram and signal space diagram. [5]
- b) Differentiate between FDMA and TDMA. Draw T1 and E1 telephone hierarchy. [3+2]
10. Why convolution coder is better than block coder? Determine systematic and non-systematic code vector for a (7,4) cyclic hamming code for message vector {1011} with generator matrix $g(x)=1+X+X^3$. [2+5]